Q1 - 24 June - Shift 1

The equations of two waves are given by :

$$y_1 = 5\sin 2\pi(x - vt)\text{cm}$$

$$y_2 = 3\sin 2\pi(x - vt + 1.5)\text{cm}$$

These waves are simultaneously passing through a string. The amplitude of the resulting wave is

(A) 2 cm (B) 4 cm

(C) 5.8 cm (D) 8 cm

Space for your notes:

Q2 - 24 June - Shift 2

Two light beams of intensities in the ratio of 9 : 4 are allowed to interfere. The .ratio of the intensity of maxima and minima will be :

(A) 2 : 3 (B) 16 : 81

(C) 25 : 169 (D) 25 : 1

Space for your notes:

Q3 - 24 June - Shift 2

Two travelling waves of equal amplitudes and equal frequencies move in opposite directions along a string. They interfere to produce a stationary wave whose equation is given by

$$y = (10 \cos \pi x \sin \frac{2\pi t}{T}) \text{ cm}$$

The amplitude of the particle at $x = \frac{4}{3}$ cm will be

________ cm.

Space for your notes:

Q4 - 25 June - Shift 1

The first overtone frequency of an open organ pipe is equal to the fundamental frequency of a closed organ pipe. If the length of the closed organ pipe is 20 cm. The length of the open organ pipe is _____ cm.

Space for your notes:

Q5 - 26 June - Shift 2

A set of 20 tuning forks is arranged in a series of increasing frequencies. If each fork gives 4 beats with respect to the preceding fork and the frequency of the last fork is twice the frequency of the first, then the frequency of last fork is_____Hz.

Space for your notes:

Q6 - 27 June - Shift 1

An observer moves towards a stationary source of sound with a velocity equal to one-fifth of the velocity of sound. The percentage change in the frequency will be :

(A) 20% (B) 10%

(C) 5% (D) 0%

Space for your notes:

Q7 - 27 June - Shift 2

If a wave gets refracted into a denser medium, then which of the following is true?

(A) wavelength speed and frequency decreases.

(B) wavelength increases, speed decreases and frequency remains constant.

(C) wavelength and speed decreases but frequency remains constant.

(D) wavelength, speed and frequency increases.

Space for your notes:

Q8 - 28 June - Shift 1

A radar sends an electromagnetic signal of electric field (E_0) = 2.25 V/m and magnetic field (B_0) = 1.5×10^{-8} T which strikes a target on line of sight at a distance of 3 km in a medium. After that, a pail of signal (echo) reflects back towards the radar vit1i same velocity and by same path. If the signal was transmitted at time t_0 from radar. then after how much time echo will reach to the radar?

(A) 2.0×10^{-5} s

(B) 4.0×10^{-5} s

(C) 1.0×10^{-5} s

(D) 8.0×10^{-5} s

Space for your notes:

Q9 - 28 June - Shift 1

The velocity of sound in a gas. in which two wavelengths 4.08m and 4.16m produce 40 beats in 12s, will be :

(A)$2.82.8\ ms^{-1}$ (B) $175.5\ ms^{-1}$
(C) $353.6\ ms^{-1}$ (D) $707.2\ ms^{-1}$

Space for your notes:

Q10 - 28 June - Shift 2

A tuning fork of frequency 340 Hz resonates in the fundamental mode with an air column of length 125 cm in a cylindrical tube closed at one end. When water is slowly poured in it, the minimum height of water required for observing resonance once again is ______cm.

(Velocity of sound in air is $340\ ms^{-1}$)

Space for your notes:

Q11 - 29 June - Shift 1

A longitudinal wave is represented by $x = 10 \sin 2\pi\left(nt - \frac{x}{\lambda}\right)$ cm. The maximum particle velocity will be four times the wave velocity if the determined value of wavelength is equal to :

(A) 2π (B) 5π
(C) π (D) $\frac{5\pi}{2}$

Space for your notes:

Q12 - 29 June - Shift 2

In an experiment to determine the velocity of sound in air at room temperature using a resonance is observed when the air column has a length of 20.0 cm for a tuning fork of frequency 400 Hz is used. The velocity of the sound at room temperature is 336 ms^{-1}. The third resonance is observed when the air column has a length of ____cm.

Space for your notes:

Answer Key

Q1 (A)	**Q2** (D)	**Q3** (5)	**Q4** (80)
Q5 (152)	**Q6** (A)	**Q7** (C)	**Q8** (B)
Q9 (D)	**Q10** (50)	**Q11** (B)	**Q12** (104)

Q1 (A)

$$A_1 = 5 \quad A_2 = 3$$

$$\Delta\theta = 2\pi(1.5) = 3\pi$$

$$A_{net} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(3\pi)}$$

$$= |A_1 - A_2|$$

$$= 2\text{cm}$$

Q2 (D)

$$\sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{9}{4}} = \frac{3}{2}$$

$$\left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}}\right)^2 = 5^2 = 25$$

Q3 (5)

$$10\cos\left(\frac{4\pi}{3}\right)$$

Q4 (80)

$$f_1 = \frac{2v}{2l_1}$$

$$f_2 = \frac{v}{4l_2}$$

$$f_1 = f_2$$

$$= \frac{2v}{2l_1} = \frac{v}{4l_2}$$

$$l_1 = 4l_2 = 80 \text{ cm}$$

Q5 (152)

$f_1 = f$

$f_2 = f + 4$

$f_3 = f + 2 \times 4$

$f_4 = f + 3 \times 4$

$f_{20} = f + 19 \times 4$

$f + (19 \times 4) = 2 \times f$

$f = 76$ Hz.

Frequency of last tuning forks $= 2f$

$= 152$ Hz

Q6 (A)

$$i = 2r$$

$$\sin i \times n_1 = \sin r \times n_2$$

$$\sin i \times 1 = \sin \frac{i}{2} \times \sqrt{2n}$$

$$\frac{\sin i}{\sin \frac{i}{2}} = \sqrt{2n}$$

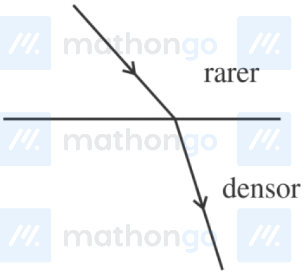
$$\frac{2 \sin \frac{i}{2} \cos \frac{i}{2}}{\sin \frac{i}{2}} = \sqrt{2n}$$

$$\cos \frac{i}{2} = \sqrt{\frac{n}{2}}$$

$$\frac{i}{2} = \cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$$

$$i = 2\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$$

Q7 (C)



No change in frequency but speed and wave-length decreases.

Q8 (B)

$$C = \frac{E_0}{B_0} = \frac{2.25}{1.5 \times 10^{-8}} = 1.5 \times 10^8 \text{ ms}^{-1}$$

$$t = \frac{6 \times 10^3}{1.5 \times 10^8} = 4 \times 10^{-5} \text{ s}$$

Q9 (D)

$$f_b = f_1 - f_2$$

$$\frac{v}{4.08} - \frac{v}{4.16} = \frac{40}{12}$$

$$\Rightarrow v = 707.2$$

Q10 (50)

Assumption : Ignore word "**fundamental mode**" in question.

$$\lambda = \frac{V}{f} = \frac{340}{340} = 1\text{m}$$

First resonating length = $\frac{\lambda}{4} = 25$ cm

Second resonating length = $\frac{3\lambda}{4} = 75$ cm

Third resonating length = $\frac{5\lambda}{4} = 125$ cm

Height of water required = 125 – 75 = 50 cm

Q11 (B)

$$V_P \max = 4V_{\text{wave}}$$

$$\omega A = 4\left(\frac{\omega}{k}\right) \Rightarrow A = \frac{4\lambda}{2\pi}$$

$$\lambda = \frac{2\pi A}{4} \Rightarrow \frac{20\pi}{4} \Rightarrow 5\pi$$

Q12 (104)

For first resonance

$$\ell_1 + e = \frac{\lambda}{4}$$

$$\lambda = \frac{336}{400} \times 100\text{ cm} = 84\text{ cm} \Rightarrow \frac{\lambda}{4} = 21\text{ cm}$$

e = 21 – 20 = 1 cm

For third resonance

$$\ell_3 + e = \frac{5\lambda}{4} = 105\text{ cm} \Rightarrow \ell_3 = 104\text{ cm}$$